SOFIUL ALAM
Senior Python Developer | AWS Serverless | Terraform | CI/CD

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Open to Hybrid / Remote | Available Immediately

PROFESSIONAL SUMMARY

A Senior Python Engineer with 25+ years' global experience leading high-performing development teams and delivering secure, mission-critical systems for UK public-sector and regulated organisations, including NHS Digital and the BBC. Skilled in delivering large-scale systems across Oil & Gas, Finance, Media, Retail, and Healthcare industries, with a strong record of accelerating business growth through technical innovation and strategic engineering.

Specialist in Python and FastAPI services on AWS – serverless Lambda functions, S3, EC2, ECS, SQS/SNS, RDS and DynamoDB – with infrastructure as code in Terraform and CloudFormation, and automated CI/CD pipelines (GitHub Actions, Azure DevOps). Also experienced in event-driven systems, ETL pipelines, cloud-native deployments on Azure, and payment integrations.

Experienced delivering within public-sector governance and security standards (NHS Digital, BBC), applying TDD, strict code review and automated deployment to ship reliable releases.

Proven track record of:

• Building idempotent distributed systems

• Building Python AWS Lambda and CloudFormation event-driven services for BBC iPlayer during Euro 2024, Olympics and General Election traffic peaks

• Cutting infrastructure provisioning time from days to minutes with Terraform (Edit Cloud)

• Scaling platforms to £1B+ revenue environments

• Led critical integrations during £2bn Virgin Media + O2 merger (Strapi + MuleSoft + payment systems)

• Turned around the entire EDIT CLOUD product delivery in 5 months in FastAPI on AWS (previous team worked 1.5+ years without go-live)

• Successfully delivered NHS Digital COVID-19 & Flu vaccination platforms on time using React.js, Python, .NET Core + full Azure stack via Terraform

• Delivered real-time well surveillance dashboards & React Native mobile apps for Kuwait Oil Company (2000+ wells)

• Reducing operational cost through automation and cloud optimisation

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CORE TECHNICAL SKILLS

Backend: Python • FastAPI • Django • Flask • AsyncIO • Celery • SQLAlchemy • Boto3, Node.js, .NetCore
Databases: PostgreSQL • Azure SQL • DynamoDB • CosmosDB • Redis, MSSQL, Oracle
Cloud: AWS (Lambda, S3, EC2, ECS/Fargate, SQS, SNS, RDS, CloudFormation, CloudWatch) • Azure (Service Bus, Functions, APIM, Logic Apps, Databricks)
DevOps: Docker • Terraform • CI/CD • GitHub Actions • Azure DevOps
Architecture: Microservices • Serverless • Event-driven systems • Idempotent processing • OAuth2/JWT • Payment Integrations (Adyen, PayPal)

WORK EXPERIENCE

Senior AI Engineer – Binary Reform (TenderAI)

May 2026 – Oct 2026

Technology Used
Python, FastAPI, LangGraph, LangChain, Celery, Redis, PostgreSQL, SQLAlchemy, Docker, AWS S3, AWS ECS/Fargate, Terraform, Boto3, OpenAI GPT, OpenCV, YOLOv8, Roboflow, Runpod, Tesseract OCR, PyMuPDF, Computer Vision APIs, Next.js, TypeScript

Key Contributions & Results
• Delivered TenderAI, an enterprise AI tender automation platform for the UK construction sector, using a LangGraph multi-agent async architecture → reduced tender preparation time by 70%.
• Built Python FastAPI endpoints for document and drawing upload, BOQ generation, job status and results retrieval, backed by Celery + Redis workers and PostgreSQL → enabled scalable asynchronous processing of large multi-format tender packs.
• Designed the multi-tenant v2 platform on AWS ECS/Fargate with all infrastructure defined in Terraform – one isolated stack per tenant, schema-per-tenant PostgreSQL and a dedicated Redis per stack → secure tenant data isolation and repeatable environment provisioning.
• Built S3-backed storage for uploaded tender packs and generated BOQ outputs, accessed from FastAPI and Celery workers via Boto3 → durable, scalable handling of large drawing sets.
• Improved BOQ pipeline accuracy with sheet-type routing, revision deduplication, floor-range multipliers and title-block extraction → more reliable quantity take-off across drawing sets.
• Engineered an isolated Door Count V2 module, evolving from OpenCV template matching to a YOLOv8 pipeline trained on Roboflow-labelled data using Runpod GPUs → automated door counting from architectural floor plans.
• Integrated Tesseract OCR, PyMuPDF, OpenAI GPT and computer vision APIs for multi-format document analysis → automated extraction of tender data from PDF and drawing inputs.
• Resolved AWS S3 storage diagnostics and containerised services with Docker → improved pipeline reliability and deployment consistency.

Lead Python Engineer – Jellycat (UK / EU / US / China)

July 2025 – Jan 2026

Technology Used
FastAPI, Azure Service Bus, Azure Logic Apps, Celery, Docker, PostgreSQL, SAP, Dynamics 365, PayPal API, Adyen API, Elee, OMS, Korber

Key Contributions & Results
• Built idempotent Python FastAPI microservices integrating Dynamics 365 ↔ OMS across 4 global regions using Azure Service Bus as message broker → improved transaction reliability and eliminated duplicate processing across £1B revenue B2C and B2B operations.
• Implemented PayPal and Adyen payment capture & refund FastAPIs with database synchronisation → reduced reconciliation errors and improved refund processing speed by 40%.
• Developed Celery-based async task runners in Azure Containers for orders, stock, stock count, stock movement, SKU, refunds → ensured high-throughput distributed processing during peak seasonal demand.
• Engineered ETL JSON → XML transformation layer for Körber ↔ ELee integrations → enabled seamless multi-system ERP synchronisation.
• Delivered end-to-end SAP ↔ Azure SQL automation via Logic Apps → removed manual finance intervention.
• Authored comprehensive FastAPI unit/integration tests → improved deployment confidence and reduced production defects.

Lead Python Engineer / Architect – Edit Cloud

Oct 2024 – Apr 2025

Technology Used
Python, FastAPI, Boto3, AWS Lambda, AWS S3, EC2, Terraform, PostgreSQL, DynamoDB, Aurora RDS, Celery, Docker, GitHub Actions CI/CD, C#

Key Contributions & Results
• Delivered full product suite in 5 months (previous team failed after 1.5 years) using FastAPI microservices on AWS → accelerated go-live and revenue generation for Video Editing & LMS cloud platform.
• Built high-throughput file ingestion FastAPI backed by AWS S3 → improved video asset upload performance by 60%.
• Built Python AWS Lambda functions triggered by S3 uploads → serverless video asset processing.
• Designed PostgreSQL + DynamoDB hybrid schema → optimised metadata and workflow processing.
• Implemented Terraform multi-cloud infrastructure → reduced provisioning time from days to minutes, including VM deployment and Adobe Premiere cloud licensing.
• Structured Terraform as reusable modules per environment and ran plan/apply through CI/CD pipelines → repeatable, reviewable infrastructure changes.
• Used Boto3 to automate S3 asset storage and EC2 provisioning from Python services → scripted, consistent cloud resource management.
• Built EC2 cost-optimisation automation → significantly reduced AWS operational costs.
• Developed C# API endpoints integrating Jira, Monday.com, Asana.

• Mentored engineers and led pair programming sessions.

Senior Python Engineer – BBC iPlayer

May 2024 – Aug 2024

Technology Used
Python, FastAPI, Django, AWS Lambda, CloudFormation, EC2, ElastiCache, CloudWatch, CI/CD, Node.js, TypeScript

Key Contributions & Results
• Developed Python event-driven microservices and AWS Lambda functions supporting Euro 2024, Olympics, Wimbledon, General Election, Glastonbury Festival → ensured stability during multi-million user peaks.
• Wrote CloudFormation templates for scalable infrastructure → enabled auto-scaling and high availability.
• Shipped infrastructure and service changes through CI/CD pipelines with CloudFormation → consistent, repeatable deployments.
• Built CloudWatch monitoring & optimisation workflows (metrics, logs, alarms) → reduced cloud over-provisioning costs.
• Used ElastiCache to cache high-traffic responses → reduced backend load during live-event traffic spikes.
• Led decommissioning of legacy services → simplified architecture.
• Built Node.js/TypeScript auto-scaling scripts for major 2024 live events.

Lead Python Engineer – Kuwait Oil Company

Oct 2022 – Nov 2023

Technology Used
Python, FastAPI, Azure Databricks, Azure Stream Analytics, PostgreSQL, CosmosDB, Docker, React, React Native, Azure SQL

Key Contributions & Results
• Developed Python FastAPI dashboards analysing 2000+ oil wells → improved operational visibility and forecasting.
• Built Invoice/Tender management backend Python Fast Api → reduced processing delays by 60%.
• Built ETL pipelines using Azure Databricks + Stream Analytics → automated geological data processing.
• Designed real-time well surveillance APIs → enhanced field decision-making speed.
• Delivered Python Fast Api backend services powering React Native field applications for Engineers to view live data.
• Mentored engineers through structured pairing sessions.

Lead Software Engineer – NHS Digital

May 2022 – Oct 2022

Technology Used
.NET Core API, React.js, Azure Functions, Terraform, Azure DevOps CI/CD, Azure SQL, REST APIs, TDD

Key Contributions & Results
• Delivered national vaccination booking backend systems → supported millions during COVID-19 & Flu rollout → Vaccinated millions of people on time and saved lives.
• Built serverless Azure Functions → elastic scaling through national demand peaks.
• Implemented TDD frameworks → reduced regression defects.
• Automated Azure infrastructure via Terraform → improved deployment reliability.
• Built Azure DevOps CI/CD pipelines running automated tests and Terraform deployments → consistent, auditable releases within NHS governance and security standards.

Lead Software Engineer / Architect – Johnson Matthey

Jan 2022 – May 2022

Technology Used
Python FastAPI, React, IoT Platforms, Azure Data Lake

Key Contributions & Results
• Spearheaded digital transformation initiatives in the Oil & Gas sector.
• Built real-time data streaming platform (Azure Data Lake + React/Python FastAPI).
• Developed IoT digital platform solutions with Python FastAPI backend services.
• Enforced strict code reviews and mentoring standards.

Lead Software Engineer – Accenture (Virgin Media + O2 Merger)

Aug 2021 – Dec 2021

Technology Used
React.js 18, TypeScript, Python FastAPI, Strapi CMS, MuleSoft, Payment Integrations, Cypress

Key Contributions & Results
• Delivered merged product catalogue & ecommerce platform (£2bn merger).
• Integrated Strapi CMS, MuleSoft, payment gateways in React + Python FastAPI.
• Implemented Cypress automation → significantly reduced manual testing time.
• Led team of 4 full-stack engineers.

Senior Software Engineer – Charles Taylor Insurtech

Jan 2021 – Aug 2021

Technology Used
WPF, Python Fast Api, AWS, Pytest, SQL Server, SSRS

Key Contributions & Results
• Built marine insurance quotation & policy systems → Reduce manual process and improve overall automated operational workflow.
• Applied TDD using Pytest → improved code quality.
• Executed database & SSRS data processing via ETL process → Build efficient and trusted data.
• Delivered CRM solutions optimising registration & exam processing workflows.

Lead Software Engineer – AAT

Dec 2019 – Dec 2020

Technology Used
AWS, React.js, Python Fast Api Microservices, CI/CD, Microsoft Dynamics CRM, BizTalk

Key Contributions & Results
• Led AWS Python fast api microservices transformation for accounting platform → Delivered LMS system for accounting and membership and increased revenue by 30%.
• Set up CI/CD pipelines for automated testing and deployment of Python microservices to AWS → faster, more reliable releases.
• Directed CRM & BizTalk integrations → Improved operational activities by 25%.
• Managed and mentored team of 8 senior developers.

Lead Software Engineer / Architect – AnaCap Private Equity

Mar 2019 – Dec 2019

Technology Used
React, Python Fast Api, AWS

Key Contributions & Results
• Delivered Minerva private equity portal React.js and Python Fast API → Invested multi million company by 25% through portal.
• Led AWS migration and rewrote the Python api single handed due to developers delivering the wrong api.
• Managed offshore team of 4 developers providing standards, strict code review and pair programming.

Lead Senior Software Engineer – Imprima (iRooms VDR)

Feb 2018 – Mar 2019

Technology Used
Angular 12, Azure Service Bus, Azure SQL, Azure Functions, Python FastAPI, CosmosDB

Key Contributions & Results
• Designed a secure virtual data room platform in Fastapi and Angular → The platform generated £9m in a single year.
• Engineered Azure-based asset management system.
• Contributed AI-driven automation initiatives → Ai driven data structure including document classification, summarization, view documents.
• Led team of 8 developers enforcing BDD & optimisation practices.

Lead Software Engineer – KPMG (HSBC Trader Surveillance)

Mar 2017 – Jan 2018

Technology Used
Python ML, React, BI Dashboards, Fast Api.

Key Contributions & Results
• Built AI-powered voice & trade surveillance system → Reduce manual process by 60%.
• Developed Python API for audio format conversion → Identify Trader misconduct and highlight Trader surveillance issues .
• Led MAGNA project & BI dashboard development → Real time surveillance reduces manual work by 60%.
• Managed cross-functional offshore team.

Principal Lead Software Engineer – Fujitsu (RBS)

Mar 2016 – Dec 2016

Technology Used
Angular, C# Web API

Key Contributions & Results
• Led team of 28 delivering £2M PMP web portal for RBS.
• Directed full SDLC and financial technology innovation.

INTERNATIONAL EXPERIENCE – QATAR

Lead Software Engineer – Qatar Gas
Built enterprise backend services improving reporting automation.
Implemented CI/CD pipelines reducing deployment cycle time.

Senior Software Engineer – RasGas
Developed APIs & ETL pipelines streamlining infrastructure reporting.
Automated database validation workflows.

Software Development Manager – Weill Cornell Medical College (Qatar)
Led hospital operational systems digitisation.
Managed secure Azure cloud deployments.

AI PROJECT (Python-Based)

Built enterprise AI tender automation system using:
Python, FastAPI, Celery, PostgreSQL, Tesseract OCR, PyMuPDF, OpenAI GPT, Computer Vision APIs

• Automated analysis of 150+ page multi-format documents → reduced tender preparation time by 95%.
• Implemented multi-agent async architecture → improved throughput.
• Reduced OpenAI API costs by 60% through token optimisation.

EDUCATION & CERTIFICATIONS

BSc Computer Science – University of Westminster
Azure Developer Associate – Firebrand
Microsoft Certified Professional

INTERESTS

Scuba Diving • Marathon Running • Digital Photography • Travel • Camping

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Reference

Available on request.